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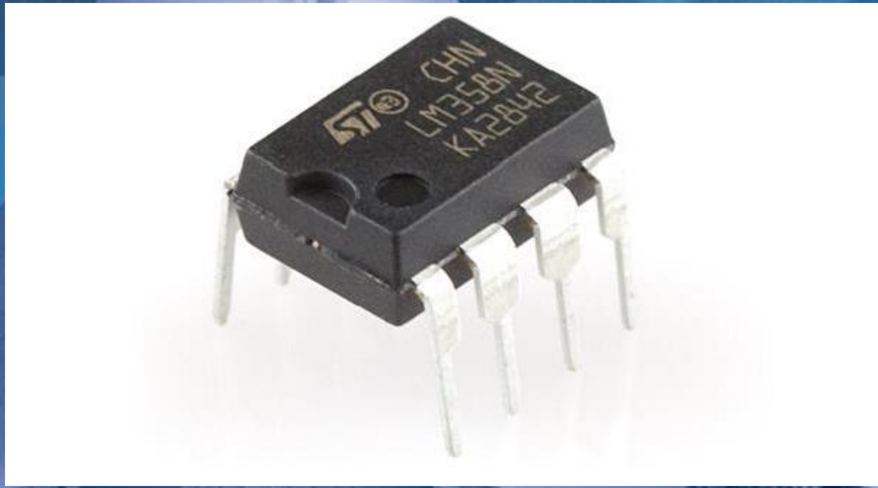
Introduction to Operational Amplifier Circuits

(EBN 122: Electricity and Electronics)



22 September 2017

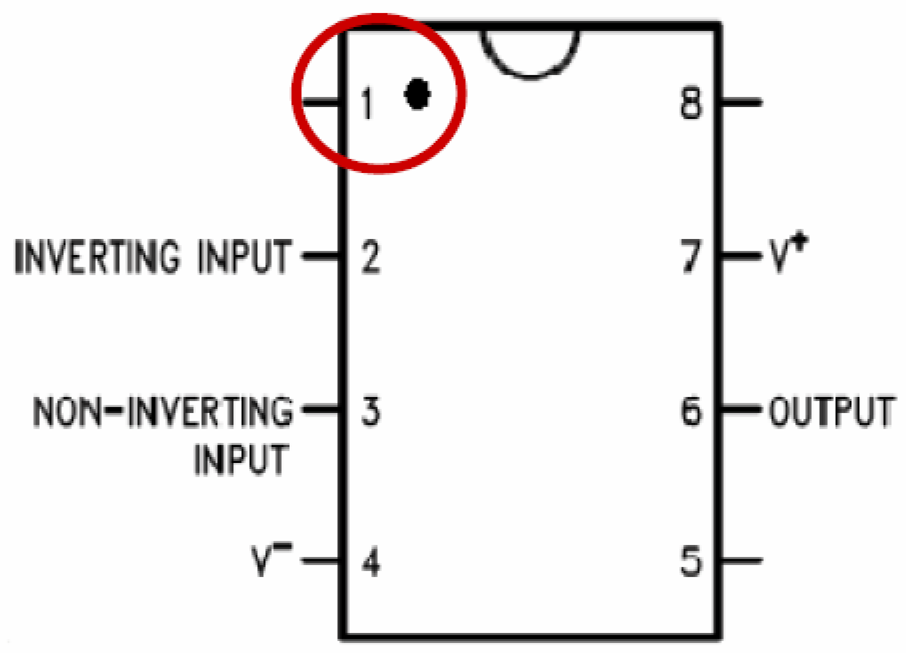
Department of Electrical, Electronic and Computer Engineering

Introduction

- Build **two Op-Amp** circuits
- Record the circuit characteristics
  - Input voltage vs Output voltage
- ***<u>Important:</u>*** Read about the basic concepts in your textbook
- During the practical you will receive the following:
  - **One** UA741 Op-amp **per group**
  - Connecting wires

Op-Amp

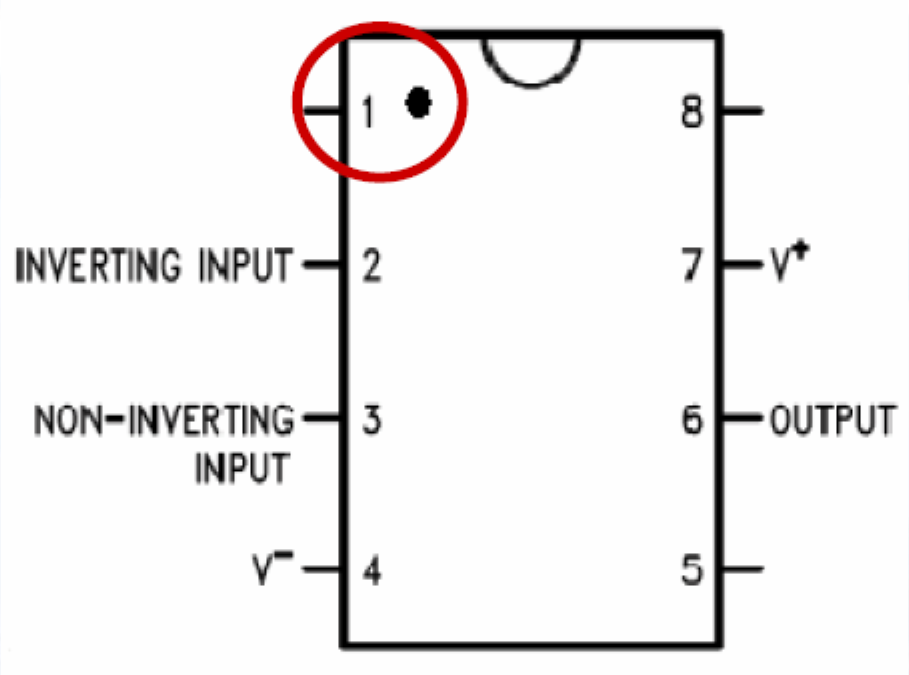
- Op-Amp: High-gain voltage amplifier with a differential input (+ and -) and usually a single-ended output.
- Used to perform mathematical operations on voltage signals.
  - Inversion
  - Addition
  - Subtraction
  - Integration
  - Etc.



Op-Amp

- Op-Amp used for this practical: 8 pin Dual In-Line Package.
- The pins correspond to the various inputs and outputs of the Op-Amp.
- **Note: Pin 1** is identified with

the **dot (encircled in red)** next to it and all other pins are **counted counter-clockwise from pin 1.**



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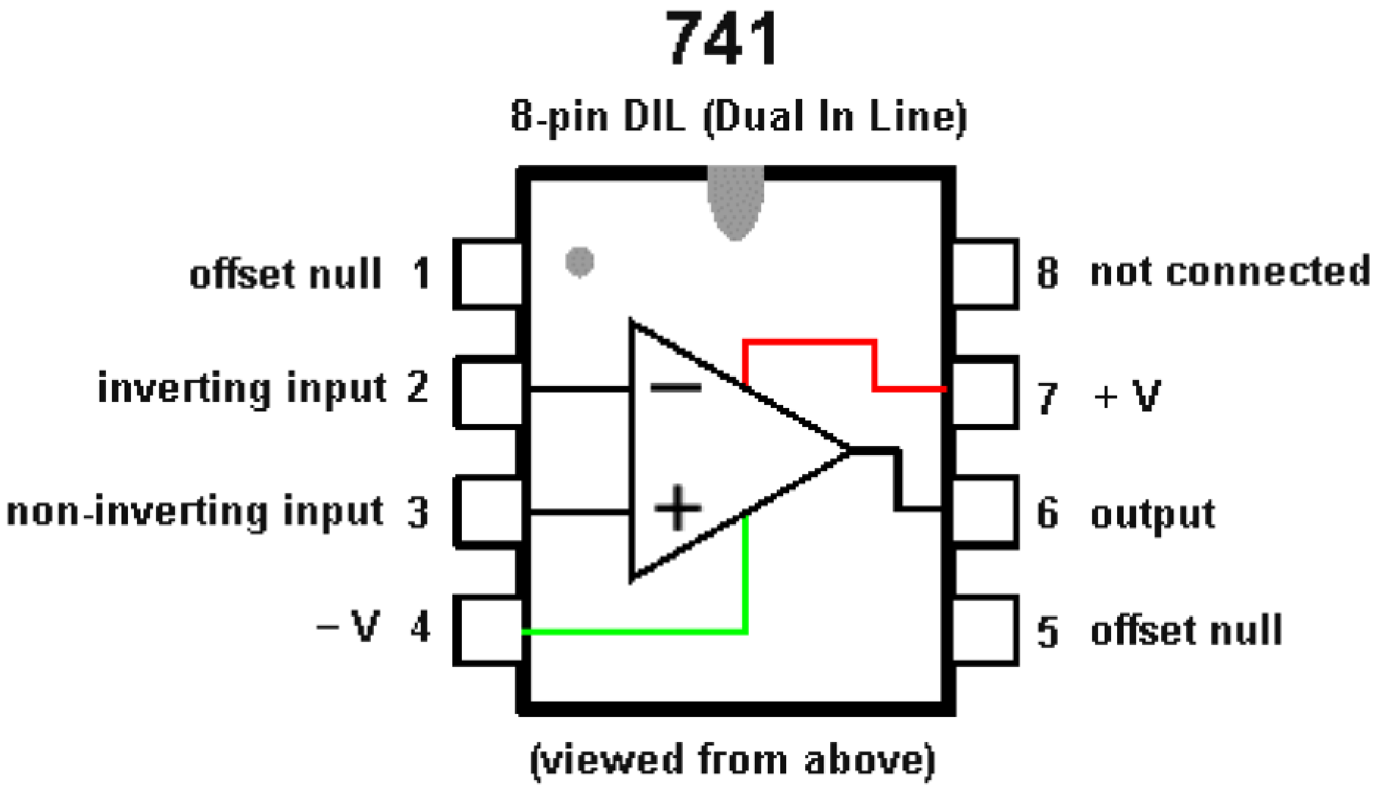
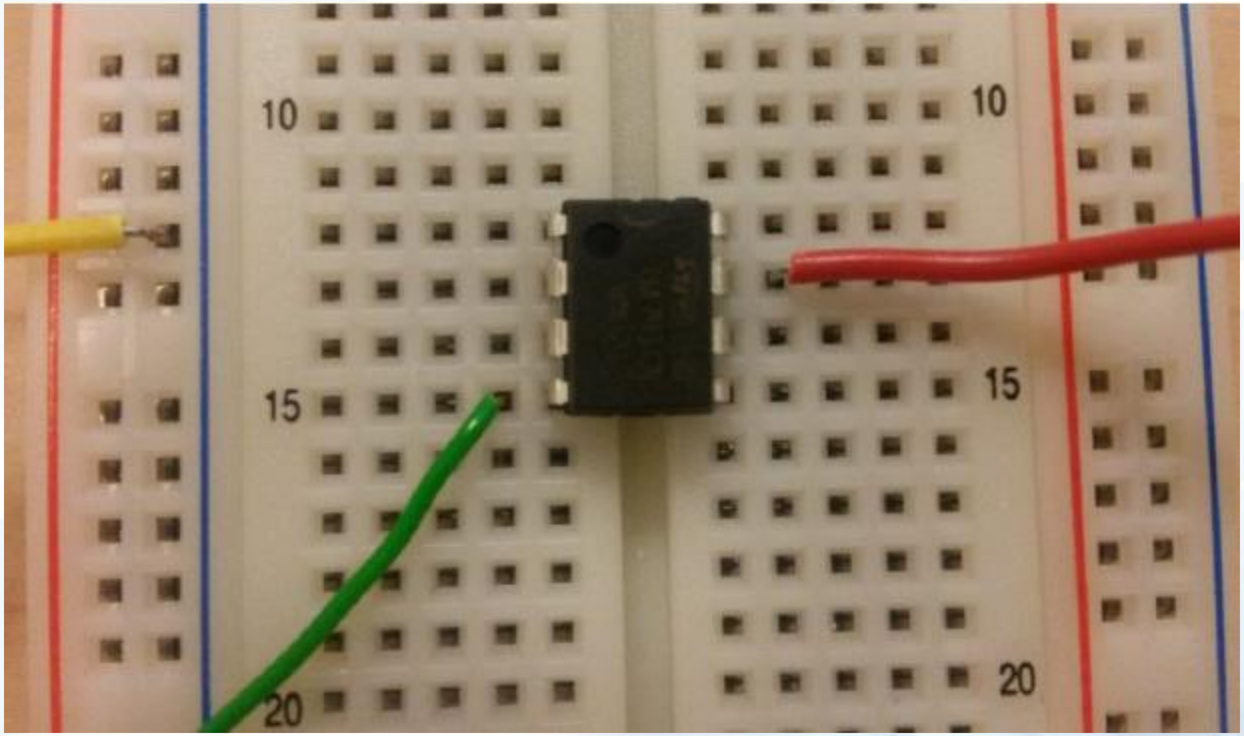
Op-Amp

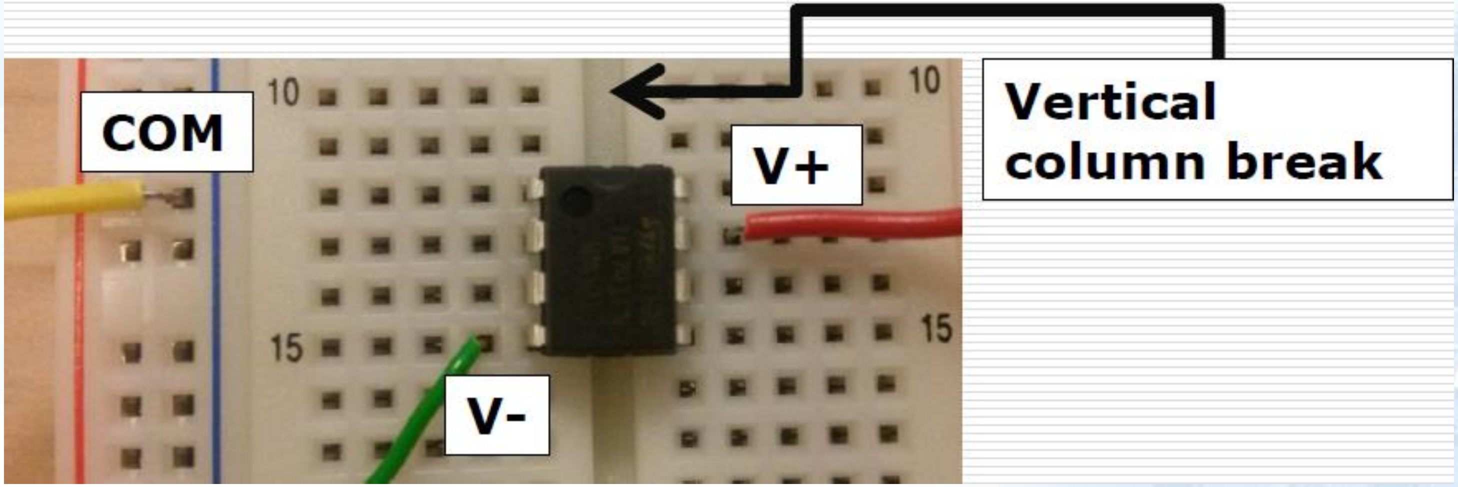

741
8-pin DIL (Dual In Line)
offset null 1
inverting input 2
non-inverting input 3
− V 4
8 not connected
7 + V
6 output
5 offset null
(viewed from above)

Constructions on Breadboard



Constructions on Breadboard

- Place the Op-Amp on the breadboard so that pin 1-4 and 5-8 are separated by the **vertical column break** on the board – ***as indicated below.***
- ***Hint:*** Connect the -12V (pin 4), +12V (pin 7) and ground (COM) rails to the breadboard as indicated.
- ***Read practical 3 guidelines for more detail.***



Workstation

Power Supply

Function generator (Oscilloscope)

Practical Preparation

- ***Important:*** Lab rules

www.up.ac.za/en/eece/article/2008415/undergraduate-laboratory-rules

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- Practical 3 **guidelines** are available on **clickUP.**

- Bring your **component boxes.**

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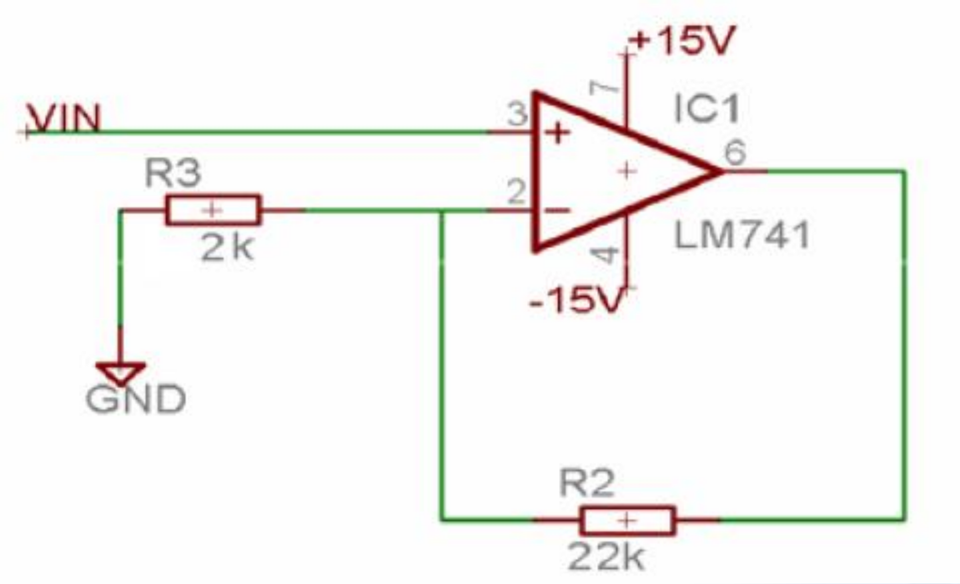
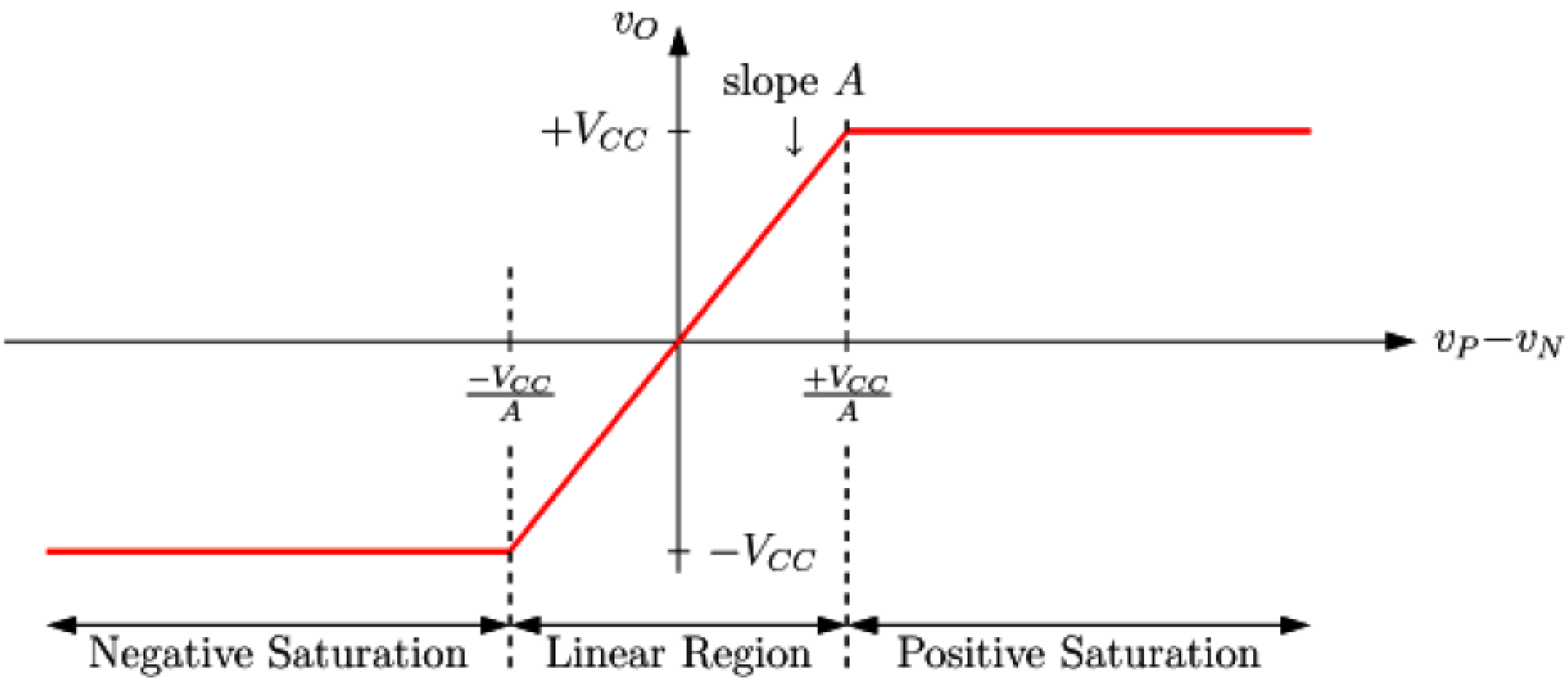
Practical Test Preparation

- You should be able to answer the following questions:
  - Is this an inverting or non-inverting amplifier?
  - Calculate the theoretical gain of the amplifier.
  - What is the relationship between input and output voltage?



Practical Test Preparation

- **Amplification and saturation regions**



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Conclusion

- Handle the Op-Amp with care
- **Please return** the Op-Amp and connecting wires after the practical
- Questions: ebn2017pracs@gmail.com

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